

Curriculum Vitae

Inhyeok Lee



Personal Data

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EDUCATION

MAR. 2021~ AUG. 2026	Ph. D. in Electrical Engineering, POSTECH	Pohang, South Korea
MAR. 2019~ FEB. 2021	M. S. in Electrical Engineering, POSTECH	Pohang, South Korea
MAR. 2015~ FEB. 2019	B. S. in Electrical Engineering, POSTECH	Pohang, South Korea

Research Interests

- Synthetic Aperture Radar (SAR) – motion compensation, resolution improvement
- Inverse Synthetic Aperture Radar (ISAR) – Complex motion, resolution improvement
- Compressed sensing (CS) – Optimization, sparse solution, SAR/ISAR imaging
- Deep learning

Publications

- *International journals*

1. In-Hyeok Lee, Oh-Tae Jang, and Kyung-Tae Kim, “Data-Driven Pulse Scoring Approach for ISAR Imaging Under Complex Rotational Motion,” (In-Preparation).
2. Oh-Tae Jang, In-Hyeok Lee, and Kyung-Tae Kim, “Consistent Target Representation Learning under Clutter Variations for SAR ATR,” *IEEE Geoscience and Remote Sensing Letters* (Submitted), June. 2026.
3. Min-Gon Cho, In-Hyeok Lee, Oh-Tae Jang, and Kyung-Tae Kim, “Diagnosing the Decision Gap in Super-Resolution-Enhanced SAR-ATR,” *IEEE Transactions on Aerospace and Electronic Systems* (Submitted), May. 2026.

4. **In-Hyeok Lee** and Kyung-Tae Kim, "Frame Scoring-Based Soft-Integration Approach for High-Resolution ISAR Imaging Under 3-D Rotational Motion," *IEEE Geoscience and Remote Sensing Letters*, Early Access, May, 2026.
5. **In-Hyeok Lee**, Min-Gon Cho, Hyun-Dong Kim, and Kyung-Tae Kim, "Efficient Compressed Sensing-Based Backprojection Approach for Small Drone-Borne W-Band SAR Imaging," *Remote sensing*, April, 2026.
6. **In-Hyeok Lee**, Min-Gon Cho, Hyun-Dong Kim, and Kyung-Tae Kim, "High-Resolution ISAR Imaging of Highly Maneuvering Targets in 3-D Rotational Motion via Frame Selection and Sparse Aperture Imaging," *IEEE Sensors Journal*, Jan. 2026.
7. Min-Gon Cho, **In-Hyeok Lee**, Hyun-Dong Kim, Oh-Tae Jang, and Kyung-Tae Kim, "Deep Learning-Based Super-Resolution of SAR Object Images via Multi-Resolution Data Augmentation," *IEEE Transactions on Geoscience and Remote Sensing* (major revision, submitted), Nov. 2025.
8. Kyung-Min Lee, **In-Hyeok Lee**, Min-Gon Cho, and Kyung-Tae Kim, "Modified Range Resolution Improvement for Specular-Reflection Bistatic Synthetic Aperture Radar Imaging," *IEEE Transactions on Aerospace and Electronic Systems*, Aug. 2023.
9. Kyung-Min Lee, **In-Hyeok Lee**, Min-Gon Cho, and Kyung-Tae Kim, "Range Resolution Improvement Using Cross-Track Interferometry for FMCW Radar," in *IEEE Geoscience and Remote Sensing Letters*, vol. 20, pp. 1-5, July, 2023.
10. Kyung-Min Lee, **In-Hyeok Lee**, and Kyung-Tae Kim, "Sparsity-driven Inverse Synthetic Aperture Radar Imaging using Accelerated Meta-Heuristic Optimization," in *IEEE Transactions on Aerospace and Electronic Systems*, Sep. 2022.
11. Bo-Hyun Ryu, **In-Hyeok Lee**, Byung-Soo Kang, and Kyung-Tae Kim, "Frame Selection Method for ISAR Imaging of 3-D Rotating Target Based on Time-Frequency Analysis and Radon Transform," in *IEEE Sensor Journal*, vol. 22, no. 20, pp. 19953-19964, Oct. 2022.

Publications

- *Domestic Journals*

1. 김현동, **이인혁**, 조민곤, 신희섭, 옥재우, 백민영, 유응노, 김경태, "바이스택 FMCW SAR 거리 해상도 향상에 관한 연구," *한국전자파학회 논문지*, vol. 36, no. 7, July. 2025.
2. **이인혁**, 조민곤, 김현동, 신희섭, 옥재우, 백민영, 유응노, 김경태, "Master-Slave 구조를 통한 바이스택 FMCW 레이다 시스템 구성 및 ISAR 영상 형성연구," *한국전자파학회 논문지*, vol. 35, no. 5, July. 2024.

3. **이인혁**, 이경민, 조민곤, 박영주, 김상원, 김경태, “단일 주파수 정현파 기반 SAR 재밍 기법에 관한 연구,” *한국전자파학회 논문지*, vol. 34, no. 11, Nov. 2023.
4. **이인혁**, 이경민, 조민곤, 박영주, 김상원, 김경태, “SAR-ATR 기반 SAR 잡음 재밍 성능 분석,” *한국전자파학회 논문지*, vol. 34, no. 7, July. 2023.
5. 이경민, **이인혁**, 홍상근, 전영일, 나인석, 김경태, “효과적인 인공위성 SAR 기반 재밍을 위한 요구 조건 분석,” *한국전자파학회 논문지*, vol. 33, no. 2, Feb. 2022.
6. 유보현, 이경민, **이인혁**, 김경태, “특징점 추출 및 매칭을 이용한 ISAR 영상의 수직-거리 스케일링 기법 연구,” *한국전자파학회 논문지*, vol. 33, no. 2, Feb. 2022.

Conferences

- *International Conferences*

1. **In-Hyeok. Lee**, Bo-Hyun Ryu, Kyung-Min Lee, Min-Gon Cho, and Kyung-Tae Kim, “Reconstruction of Fine Cross-Range Resolution ISAR Images of Targets in 3-D Motion using Compressed Sensing and Frame Selection,” in *2023 International RADAR conference*, Sydney, Australia, Nov. 2023.
2. Kyung-Min Lee, **In-Hyeok Lee**, Min-Gon Cho, and Kyung-Tae Kim, “Two Dimensional Resolution Improvement for FMCW Synthetic Aperture Radar Using Multistatic Configuration,” in *2023 International RADAR conference*, Sydney, Australia, Nov. 2023.

Project Experiences

1. SAR 재밍기법 기술 동향 및 구현방안 연구, LIGNex1, 2021. 05 ~ 2021. 12
2. 위성SAR 전자전 모의환경 분석 모델링 연구, ADD(Agency for Defense Development), 2021. 11 ~ 2023. 11.
3. 스마트 모니터링을 위한 차세대 멀티스테틱 레이더 영상 시스템 연구, IITP(Institute of Information & Communications Technology Planning & Evaluation), 2019. 04 ~ 현재
4. W-Band 기반 FMCW 바이스테틱 SAR 영상 형성을 위한 신호 동기,

LIGNex1, 2022. 06 ~ 2025. 05

5. 비기계식 Lidar system simulation, 삼성전자, 2019. 03 ~ 2020. 02
6. 다목적실용위성 SAR 영상데이터의 초해상화 영상 변환 시스템 개발, KARI(Korea Aerospace Research Institute), 2022. 05 ~ 2023. 12

Patents

1. 김경태, **이인혁**, “항법 센서가 미장착된 드론 환경에서 주파수 변조 연속파형을 이용한 합성 개구면 레이더 영상 생성 장치 및 그 방법,” 등록번호 10-2738281, 2024년 11월 29일.
2. 김경태, **이인혁**, 이경민, “멀티스테틱 주파수 변조 연속파형 드론 기반 간섭 합성 개구면 레이더를 이용한 해상도 향상 장치 및 방법, 출원번호 10-2025-0012944, 2025년 02월 03일.
3. 홍상근, 전영일, 김경태, 이경민, **이인혁**, “합성 개구면 레이더 대응 다중허위표적 재밍신호 생성 방법,” 등록번호 10-2750684, 2025년 01월 02일.